

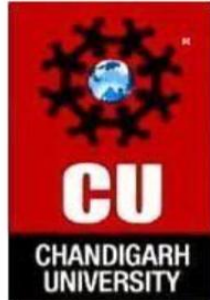


**DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING**

Discover. Learn. Empower.

**NAAC
GRADE A+**

Accredited University



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

ASSIGNMENT - 02

Subject: Computer Networks

Subject Code: 24CST-339

Student Name: Diwansh
Branch: CSE - AIML
Semester: 5

UID: 24BAI70294
Section/Group: 24AIT_KRG 1B
Submitted to- Alok Sir

Q1. What is an IP Address? Why do we need classes in case of IPV4 address? Explain each class, and mention how many host and networks are possible in each class. Also explain what is sub-netting and super-netting with examples.

Ans

IP Address

An **Internet Protocol (IP) Address** is a unique numerical address assigned to every device connected to a network. It helps in identifying devices and enables communication between them over the Internet or a local network.

IPv4 addresses are **32 bits long** and are represented in dotted decimal notation.

Example: 192.168.1.10

An IP address consists of:

- **Network ID** – Identifies the network.
- **Host ID** – Identifies the device within the network.

Need for Classes in IPv4

Different organizations require different numbers of hosts. A small company may need only a few hundred hosts, while a large organization may require millions. To efficiently allocate IP addresses and reduce wastage, IPv4 addresses were divided into different classes.

The classful addressing scheme:

- Simplifies address allocation.
- Supports networks of different sizes.
- Makes routing easier.

IPv4 Address Classes

Class A

- First Bit: **0**
- Range: **1.0.0.0 – 126.255.255.255**
- Default Subnet Mask: **255.0.0.0 (/8)**
- Network Bits: **8**
- Host Bits: **24**

Number of Networks:

$$2^7 = 128 \approx 126 \text{ usable networks}$$

Hosts per Network:

$$2^{24} - 2 = 16,777,214$$

Used for: Very large organizations.

Example: 10.20.30.40

Class B



- First Bits: **10**
- Range: **128.0.0.0 – 191.255.255.255**
- Default Mask: **255.255.0.0 (/16)**
- Network Bits: **16**
- Host Bits: **16**

Number of Networks:

$$2^{14} = 16,384$$

Hosts per Network:

$$2^{16} - 2 = 65,534$$

Used for: Medium-sized organizations.

Example: 172.16.10.20

Class C

- First Bits: **110**
- Range: **192.0.0.0 – 223.255.255.255**
- Default Mask: **255.255.255.0 (/24)**
- Network Bits: **24**
- Host Bits: **8**

Number of Networks:

$$2^{21} = 2,097,152$$

Hosts per Network:

$$2^8 - 2 = 254$$

Used for: Small organizations.

Example: 192.168.1.10

Class D

- First Bits: **1110**
- Range: **224.0.0.0 – 239.255.255.255**
- Used for **Multicasting**.
- No network-host division.

Class E

- First Bits: **1111**
- Range: **240.0.0.0 – 255.255.255.255**
- Reserved for **research and experimental purposes**.

Summary Table

Class	Range	Default Mask	Networks	Hosts/Network
A	1–126	255.0.0.0	126	16,777,214
B	128–191	255.255.0.0	16,384	65,534
C	192–223	255.255.255.0	2,097,152	254

Class Range Default Mask Networks Hosts/Network

D 224–239 Multicasting - -

E 240–255 Experimental - -

Subnetting

Subnetting is the process of dividing a large network into smaller logical networks called subnets by borrowing bits from the host portion.

Advantages

- Better IP address utilization.
- Improved security.
- Reduced network traffic.
- Easier management.

Example

Network:

192.168.1.0/24

Borrow 2 host bits:

192.168.1.0/26

Number of subnets:

$$2^2 = 4$$

Hosts per subnet:

$$2^6 - 2 = 62$$

Thus, one network is divided into four smaller subnetworks.

Supernetting

Supernetting is the process of combining multiple smaller networks into a larger network. It is also called **route aggregation** or **route summarization**.

Advantages

- Reduces routing table size.
- Improves routing efficiency.
- Simplifies network management.

Example

Four Class C networks:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

can be combined as:

192.168.0.0/22

Instead of maintaining four routes, the router maintains only one route.

Q2. What is the difference between Link state and Distance vector routing? Explain how these two algorithms help in identifying the least cost path with the help of examples.

Ans

Introduction

Routing is the process of finding the best path for transmitting data packets from a source to a destination in a network. Routers use routing algorithms to determine the most efficient path. Two important routing algorithms are **Distance Vector Routing** and **Link State Routing**. Both algorithms aim to find the **least-cost (shortest) path**, but they differ in the way they collect network information and calculate routes.

Distance Vector Routing

Distance Vector Routing is based on the **Bellman-Ford Algorithm**. In this method, each router maintains a routing table containing:

- Destination Network
- Distance (Cost)
- Next Hop Router

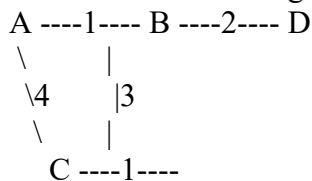
Routers periodically exchange their routing tables with neighboring routers and update their information accordingly.

Working

1. Each router knows the distance to its directly connected neighbors.
2. Routers share their routing tables with neighbors.
3. Each router updates its table if a shorter path is found.
4. The process continues until all routers have consistent routing information.

Example

Consider the following network:



Link Costs:

- $A \rightarrow B = 1$
- $A \rightarrow C = 4$
- $B \rightarrow C = 3$
- $B \rightarrow D = 2$
- $C \rightarrow D = 1$

To reach D from A:

Path 1: $A \rightarrow B \rightarrow D$

Cost = $1 + 2 = 3$

Path 2: $A \rightarrow C \rightarrow D$

Cost = $4 + 1 = 5$

Distance Vector selects:

$A \rightarrow B \rightarrow D$

Total Cost = 3

because it has the lowest cost.

Advantages

- Simple implementation.
- Requires less memory.
- Suitable for small networks.

Disadvantages

- Slow convergence.
- Routing loops may occur.
- Count-to-infinity problem.

Protocol Example

RIP (Routing Information Protocol) uses Distance Vector Routing.

Link State Routing

Link State Routing is based on **Dijkstra's Shortest Path Algorithm**.

In this approach, each router builds a complete map (topology) of the network and independently calculates the shortest path to every destination.

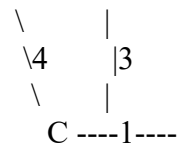
Working

1. Each router discovers its neighbors.
2. It measures the cost of each link.
3. Link State Advertisements (LSAs) are flooded throughout the network.
4. Every router receives the same topology information.
5. Each router runs Dijkstra's Algorithm to compute shortest paths.

Example

Using the same network:

A ---1--- B ---2--- D



Router A creates the topology map and calculates:

Path	Cost
$A \rightarrow B \rightarrow D$	3
$A \rightarrow C \rightarrow D$	5
$A \rightarrow B \rightarrow C \rightarrow D$	5

Shortest path:

$A \rightarrow B \rightarrow D$

Cost = 3

Thus, Dijkstra's Algorithm identifies the least-cost route.

Advantages

- Faster convergence.
- No routing loops after convergence.
- Better scalability.

- More accurate routing decisions.

Disadvantages

- Requires more memory and CPU.
- More complex implementation.

Protocol Example

OSPF (Open Shortest Path First) uses Link State Routing.

Difference Between Distance Vector and Link State Routing

Feature	Distance Vector Routing	Link State Routing
Algorithm	Bellman-Ford	Dijkstra
Network Knowledge	Knows only neighbors	Knows complete network topology
Information Sharing	Entire routing table shared	Link state information shared
Convergence Speed	Slow	Fast
Routing Loops	Possible	Rare
Memory Requirement	Low	High
CPU Requirement	Low	High
Scalability	Small networks	Large networks
Example Protocol	RIP	OSPF

How They Identify the Least-Cost Path

Distance Vector

Uses the Bellman-Ford equation:

$$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$$

Where:

- $D_x(y)$ = Cost from router x to destination y
- $c(x, v)$ = Cost to neighbor v
- $D_v(y)$ = Neighbor's cost to destination

Each router chooses the route with the minimum total cost.

Link State

Uses Dijkstra's Algorithm:

1. Build complete network graph.
2. Start from source router.
3. Select nearest unvisited node.
4. Update path costs.
5. Repeat until all nodes are visited.

The route with the smallest accumulated cost becomes the shortest path.

Example Comparison

Network:

A ----2---- B ----3---- D
 \ /

\-----7-----/

Possible paths from A to D:

1. $A \rightarrow D = 7$
2. $A \rightarrow B \rightarrow D = 2 + 3 = 5$

Both Distance Vector and Link State algorithms calculate:

Least-Cost Path = $A \rightarrow B \rightarrow D$

Total Cost = 5

The difference lies in how they obtain the information required to make the decision.

Q3. What is congestion in a network? Why do we need to control congestion? Explain the difference between Leaky bucket and Token bucket with examples. Also, explain why do we need to do fragmentation?

Ans

Congestion in a Network

Congestion occurs when the number of packets entering a network exceeds the network's capacity to process and transmit them. As a result, routers and switches become overloaded, causing delays and packet loss.

Example

If a router can handle **1000 packets per second** but receives **1500 packets per second**, the extra packets are queued or dropped, leading to congestion.

Effects of Congestion

- Increased delay
- Packet loss
- Reduced throughput
- Poor network performance

Need for Congestion Control

Congestion control is used to regulate traffic and prevent network overload.

Reasons for Congestion Control

1. Improves network efficiency.
2. Reduces packet loss.
3. Minimizes transmission delays.
4. Ensures fair bandwidth usage.
5. Maintains Quality of Service (QoS).

Leaky Bucket Algorithm

The **Leaky Bucket Algorithm** controls traffic by sending packets at a constant rate regardless of how fast they arrive.

Working

- Packets enter a bucket (buffer).
- Packets leave at a fixed rate.
- If the bucket becomes full, extra packets are discarded.

Example

If the bucket capacity is 10 packets and packets leave at 2 packets/sec, a sudden arrival of 15 packets will cause 5 packets to be dropped.

Advantages

- Simple implementation.
- Produces smooth traffic flow.

Disadvantage

- Does not allow burst traffic.

Token Bucket Algorithm

The **Token Bucket Algorithm** allows burst traffic while controlling the average transmission rate.

Working

- Tokens are generated and stored in a bucket.
- A packet can be sent only if a token is available.
- Unused tokens accumulate and can be used later.

Example

If 5 tokens are generated per second and 20 tokens are stored, a burst of 15 packets can be transmitted immediately using the accumulated tokens.

Advantages

- Allows burst transmission.
- Better bandwidth utilization.

Disadvantage

- More complex than Leaky Bucket.

Difference Between Leaky Bucket and Token Bucket

Feature	Leaky Bucket	Token Bucket
Output Rate	Fixed	Variable
Burst Traffic	Not Allowed	Allowed
Excess Traffic	Discarded	Can be sent if tokens exist
Flexibility	Less	More
Traffic Flow	Smooth	Burst-capable

Fragmentation

Fragmentation is the process of dividing a large IP packet into smaller packets so that it can pass through networks with smaller **MTU (Maximum Transmission Unit)** sizes.

Example

A packet of **3000 bytes** passing through a network with an MTU of **1000 bytes** will be divided into three fragments of approximately 1000 bytes each.

Need for Fragmentation

1. Different networks have different MTU sizes.
2. Enables communication across heterogeneous networks.
3. Prevents large packets from being discarded.
4. Ensures successful delivery of data.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Discover. Learn. Empower.



Accredited University